

MATH 1210 Worksheet 2 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

1. Write the following complex numbers in Cartesian form.

(a) $(2 + 4i) - (3 - 2i)$

(b) $\overline{2 + i} - (3 + 4i)$

(c) $\frac{1}{i}$

(d) $(1 + 2i)^2$

(e) $i^3 - 3i^2 + 2i + 4$

(f) $(1 + i)^2 + \overline{(1 + i)^2}$

(g) $\frac{(1+i)^2(2-i)}{(3+2i)^2}$

(h) $\overline{(4 - i)^2}$

(i) $\frac{1}{2i} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$

(j) $\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)^4 \left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right)^4$

2. Express the following complex numbers in polar form. Use the principal value for the argument.

(a) $-1 + i$

(b) $-2i$

(c) $\sqrt{3} - i$

(d) $-\sqrt{3} - 3i$

(e) $-2[\cos(\pi/3) - \sin(\pi/3)i]$

3. Find the Cartesian form of

$$z = \frac{i^{62}(\sqrt{2} + \sqrt{6}i)^8}{2^8(-\sqrt{6} - \sqrt{2}i)}.$$

Simplify as much as possible.

4. Find, in simplified form, the complex number z that satisfies

$$(1 - i)^2(1 + i)^2 + \overline{(4 - 3i)}z = 1 + 4i.$$

5. Solve the following equations:

(a) $x^2 + 2x + 7 = 0$,

(b) $x^4 + 4x^2 + 3 = 0$.

6. Consider the quadratic equation $x^2 - tx + 1 = 0$ where t is a real number. For what values of t does the equation have two distinct complex conjugate solutions?

7. Verify the following properties for complex numbers.

(a) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$ for two complex numbers $z_1 = a + ib$ and $z_2 = x + iy$.

(b) $\frac{z}{|z|}$ has modulus 1.

(c) Let α be a nonzero real number. Then $\arg(\alpha z) = \arg(z) + 2k\pi$ if α is positive, and $\arg(\alpha z) = \pi + \arg(z) + 2k\pi$ if α is negative.